# Smart Grain Storage Condition Monitoring and Spoilage Risk Prediction for Public Distribution System (PDS) Using AI and IoT

Gokula Kannan M
Department of AIML
Vels Institute of Science, Technology & Advanced Studies (VISTAS)
Chennai, India
gokulakannan1719@gmail.com

Umesh R
Department of AIML
Vels Institute of Science, Technology & Advanced Studies (VISTAS)
Chennai, India
.

Vignesh S
Department of AIML
Vels Institute of Science, Technology & Advanced Studies (VISTAS)
Chennai, India
.

Ms. S. J. Kavitha, Assistant Professor
Department of Computer Science and Engineering
Vels Institute of Science, Technology & Advanced Studies (VISTAS)
Chennai, India
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***Abstract*—Food grain spoilage during storage is a major challenge in the Public Distribution System (PDS). High temperature, humidity, excess moisture and poor air quality reduce grain quality and lead to food wastage and economic loss. This paper presents a Smart Grain Storage Condition Monitoring and Spoilage Risk Prediction System that uses Artificial Intelligence (AI) and the Internet of Things (IoT) to monitor storage conditions continuously and to give early warning before spoilage occurs. An ESP32 microcontroller is interfaced with a DHT22 temperature and humidity sensor, a grain moisture sensor and an MQ135 gas sensor to collect real-time environmental data from the storage area. The data is sent to an IoT cloud platform for continuous monitoring and analysis. An AI model processes the sensor data and classifies the storage condition as Safe, Moderate Risk or High Risk, so that warehouse authorities can take preventive action in time. The proposed solution aims to reduce food wastage, improve grain storage efficiency, minimise manual inspection and support smart warehouse management in PDS godowns through a low-cost, scalable and intelligent monitoring system.***

***Keywords*—Smart grain storage; Public Distribution System; Internet of Things; Artificial Intelligence; Spoilage risk prediction; ESP32***

## INTRODUCTION

The Public Distribution System (PDS) depends on warehouses to hold essential food grains such as rice, wheat and dhal for long periods before they reach the public. During this time the grain is exposed to the conditions inside the storage area, and its quality depends largely on how well those conditions are controlled.

High temperature, excessive humidity, increased grain moisture and poor air quality are the main environmental factors that harm stored grain. Together they can lead to fungal growth, insect infestation, bad odour and, eventually, spoilage. The result is food wastage, reduced grain quality and economic loss, which is a serious concern for a system that exists to supply food to the public.

In most storage facilities the condition of the grain is still checked by manual inspection. Workers examine the stock periodically, which takes time and cannot follow the conditions continuously. Temperature and humidity are not tracked all the time, gases released by spoiling grain cannot be detected in real time, and there is no automatic alert or prediction. As a result, spoilage may be noticed only after it has already started.

Internet of Things (IoT) technology makes it possible to measure storage conditions continuously with low-cost sensors and to send the readings to a remote dashboard. Artificial Intelligence (AI) can then be applied to these readings to recognise combinations of conditions that point towards deterioration, instead of relying on a single sensor value. Combining the two gives warehouse staff both a live view of the storage area and an early indication of risk.

This paper presents a Smart Grain Storage Condition Monitoring and Spoilage Risk Prediction System for PDS warehouses. An ESP32 microcontroller collects temperature, humidity, grain moisture and gas readings from a DHT22 sensor, a grain moisture sensor and an MQ135 gas sensor. The readings are transmitted over Wi-Fi to a cloud server and a monitoring dashboard, where an AI model classifies the storage condition as Safe, Moderate Risk or High Risk. Alerts are given through an LCD display, LED indicators, a buzzer and the web dashboard so that preventive action can be taken before the grain is damaged.

The objectives of the work are to monitor temperature, humidity and grain moisture inside the storage area, to detect harmful gases using the MQ135 sensor, to send the sensor data to an IoT dashboard, to predict the grain storage condition using AI, and to generate Safe, Moderate Risk and High Risk alerts, with the overall aim of reducing grain spoilage and food wastage in PDS warehouses. The system is intended for the storage of rice, wheat and pulses and is built as a low-cost prototype.

## LITERATURE REVIEW

[1]. Applications of new technologies for monitoring and predicting grains quality stored: Sensors, Internet of Things, and Artificial Intelligence. Everton Lutz and Paulo Carteri Coradi (2022) reviewed sensor-based monitoring, IoT and wireless sensor network architectures, and machine-learning approaches used for stored-grain quality. The variables considered include moisture, temperature, relative humidity and intergranular CO. The review explains how real-time

sensing and prediction can support early detection of deterioration and better storage decisions, and it provides the technical basis for combining sensors, IoT and AI in grain storage. Being a review, it does not present a field-validated PDS system, and no single integrated spoilage-risk model designed for Indian PDS warehouses is described.

[2]. Automated food grain monitoring system for warehouse using IOT. Lydia J., Leones Sherwin Vimalraj S., Monisha R. and R. Murugan (2022) developed an IoT-based warehouse monitoring system that uses sensor nodes, a PIC microcontroller and GSM or cloud connectivity, with readings reported through mobile and SMS messages. The system monitors grain level, temperature, humidity, vibration, CO, motion and smoke. It supports remote monitoring and reduces the dependence on manual inspection. However, the monitoring is mainly threshold-based, predictive estimation of spoilage risk is limited, and validation over a long duration in large PDS warehouses is absent.

[3]. Prototype wireless sensor network and Internet of Things platform for real-time monitoring of intergranular equilibrium moisture content and predict the quality corn stored in silos bags. Paulo Carteri Coradi et al. (2022) built a wireless sensor network and IoT platform for real-time temperature and relative-humidity monitoring, together with neural-network models that predict the physical, physicochemical and microbiological quality of stored corn. The work links environmental measurements to grain-quality prediction and shows the advantage of continuous monitoring over occasional sampling. It is limited to corn in bag silos and a small storage setup, so validation across PDS grains, warehouse layouts and Indian climatic conditions is still needed.

[4]. Real-time equilibrium moisture content monitoring to predict grain quality of corn stored in silo and raffia bags. Everton Lutz et al. (2022) combined a wireless sensor network and an IoT platform for real-time temperature and relative-humidity monitoring, and used artificial neural networks to predict grain quality under different moisture, temperature and packaging conditions. Their results show that equilibrium moisture content is an important indicator of quality deterioration and that real-time prediction is possible. The experiments were carried out under controlled conditions and mainly on corn, so a multi-grain dataset from real warehouses would strengthen a PDS spoilage model.

[5]. Monitoring of intergranular variables for predicting technical breakage of wheat grains stored in vertical silos. Marisa Menezes Leal et al. (2023) monitored temperature, relative humidity and CO with sensors, and applied artificial neural networks and multiple linear regression to predict dry-matter loss in stored wheat. The study shows that changes in intergranular temperature, humidity and CO give early information about changes in grain quality and can support predictive decisions. It concentrates on vertical-silo wheat storage and dry-matter loss rather than full spoilage classification, and pest, fungal and operational variables are not fully integrated.

[6]. Smart food grain storage system using Internet of Things (IoT): A Review. Supriya Priyadarsani, Asit Kumar Pradhan and Prakash Chandra Jena (2023) reviewed IoT applications for food-grain storage in the Indian context, including real-time monitoring, early warning, environmental control, inventory, traceability and predictive analytics. The paper connects IoT-based storage technology directly with Indian food-security concerns such as spoilage, mold and insect infestation. As a review paper, it does not experimentally validate a complete AI and IoT warehouse system or report predictive performance on PDS datasets.

[7]. Predicting early mycotoxin contamination in stored wheat using machine learning. Yonggik Kim, Seokho Kang, Oladayo Solomon Ajani, Rammohan Mallipeddi and Yushin Ha (2024) used a Compact Convolutional Transformer to classify wheat as healthy, incipient or contaminated, based on CO respiration and visual mold information. The reported accuracy was 83.33%. The work demonstrates the value of AI-based early classification and of separating early-stage spoilage from grain that is already contaminated. It relies on a relatively small laboratory image dataset collected under controlled conditions, and more varied field data with sensor and IoT integration are needed for PDS deployment.

[8]. Integration of dual band radio waves and ensemble-based approach for rice moisture content determination and localisation. Noraini Azmi et al. (2024) used 2.4 GHz and 868 MHz wireless measurements with an ensemble learning approach to estimate the moisture content of rice and to localise wet spots, reporting an ensemble accuracy of about 94.8%. The method supports non-invasive moisture estimation and the localisation of areas that may spoil. Its validation was carried out on a testbed, and large heterogeneous PDS warehouses, different grain types and changing radio-frequency environments need further validation.

[9]. Prototype of Smart Grain Storage Based Internet of Things. Wahyu Adi Pratama, Nurchim and Bondan W. Pamekas (2024) followed a design-based research approach to build a prototype using an ESP32, DHT22 temperature and humidity sensors, an LDR, soil-moisture sensing and cloud data transmission with early warnings. The work provides a low-cost IoT architecture for real-time grain-storage monitoring and early warning. It mainly monitors conditions; automatic control and mobile integration were left as future improvements, and AI-based spoilage-risk prediction is not its main focus.

[10]. Monitoring and predicting corn grain quality on the transport and post-harvest operations in storage units using sensors and machine learning models. Dágila Melo Rodrigues et al. (2024) monitored temperature, relative humidity and CO during transport, drying and storage, and compared regression, artificial neural network, decision tree and random forest models for predicting grain quality. The study integrates sensor measurements across the post-harvest stages. It is focused on corn and on one specific workflow, whereas a PDS implementation needs continuous warehouse data, multi-grain validation, alerts, traceability and operational integration.

The reviewed work shows that sensing, IoT and AI have all been applied to stored grain, but mostly for maize or corn, in controlled setups, or with advanced research technologies such as electronic nose systems. Several systems only monitor conditions and do not predict spoilage risk, and the predictive studies are rarely tied to a low-cost, alert-based system suited to PDS warehouses. This gap motivates the

ESP32-based AI and IoT prototype presented in this paper, which monitors temperature, humidity, grain moisture and air quality, classifies the storage condition as Safe, Moderate Risk or High Risk, and shows real-time alerts on a dashboard.

## METHODOLOGY

The proposed system monitors the storage conditions of rice, wheat and pulses in a PDS warehouse continuously and converts the sensor readings into a spoilage-risk level. The methodology has seven stages: grain storage monitoring, sensor data collection, ESP32 processing, IoT data transmission, AI analysis, spoilage risk prediction and alert generation, as shown in Fig. 1.

### *System Architecture*

In the sensing stage, three sensors cover the main factors that affect stored grain. The DHT22 measures temperature and humidity, the grain moisture sensor measures the moisture content of the grain, and the MQ135 detects air quality and harmful gases. The ESP32 microcontroller acts as the gateway. It collects the data from all the sensors, performs initial processing and filtering, and prepares the data for transmission.

The processed data is sent over Wi-Fi to a cloud server and a monitoring dashboard for real-time access. The cloud stores the readings in a real-time database, and the stored data is passed to the AI model.

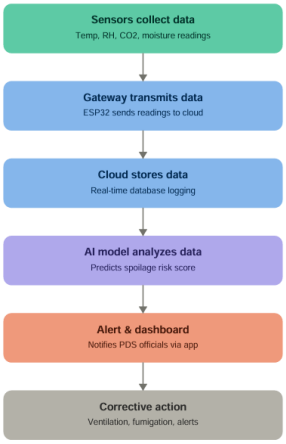



*Fig. 1. Overall system architecture of the Smart Grain Storage system, from sensor data collection to corrective action.*

A Python-based AI model analyses the incoming sensor data and identifies patterns associated with grain deterioration and spoilage. The model is trained on a dataset of temperature, humidity, gas and grain moisture values collected from grain storage conditions. Each record in the dataset is labelled as Safe, Warning or Danger according to grain quality. Live readings from the ESP32 are compared with the trained dataset to produce a prediction, and the model also gives a recommendation such as drying the grains, improving ventilation or inspecting the storage.

The AI model classifies the storage condition into three risk levels. Safe indicates a normal storage condition, Moderate Risk means that attention is required, and High Risk means that immediate action is required. Based on the predicted level, alerts are generated through the LCD display, the LED indicators, a buzzer alarm and the web dashboard, which notifies the PDS officials. The corrective actions that follow include ventilation, fumigation and further alerts. A mobile application for the same purpose is planned as a future enhancement.

## IMPLEMENTATION

The system was implemented as a working prototype in the form of a small storage unit, shown in Fig. 2. The unit has a storage chamber with a transparent door, a 16×2 LCD on the front panel that displays the sensor values and risk status, three LED indicators labelled Safe, Warning and Danger, and a buzzer for the alert. The front panel also lists the functions of the system: temperature monitoring, humidity monitoring, grain moisture detection, harmful gas monitoring, AI-based analysis and the IoT dashboard.




*Fig. 2. Prototype of the Smart Grain Storage system using IoT and AI.*

The hardware components and their roles are listed in Table I. The components are connected on a breadboard with jumper wires, and the unit is powered from a 5 V power bank, which makes the prototype portable.

TABLE I. HARDWARE COMPONENTS AND THEIR ROLES

| Component | Role in the system |
|---|---|
| ESP32 development board | Main microcontroller |
| DHT22 sensor | Temperature and humidity monitoring |
| MQ135 sensor | Gas and air-quality monitoring |
| Grain moisture sensor | Grain moisture detection |
| 16×2 LCD (I2C) | Display of values and risk status |
| Green, yellow and red LEDs | Low, medium and high risk indication |
| Buzzer | High-risk alert |
| Breadboard and jumper wires | Circuit connections |
| Power bank (5 V) | Portable power supply |

The software tools used in the project are listed in Table II. The ESP32 is programmed in the Arduino IDE, which is also used for sensor integration and data reading. The AI model is developed in Python, and Google Colab is used for dataset preprocessing, model training and evaluation. The sensor dataset is stored in Excel or CSV format, and the dashboard is a web-based interface built with CSS and JavaScript.

TABLE II. SOFTWARE TOOLS AND THEIR USAGE

| Software | Usage in the project |
|---|---|
| Arduino IDE | ESP32 programming and sensor integration |
| Python | AI model development for spoilage prediction |
| Google Colab | Dataset preprocessing and AI training |
| Web dashboard (CSS, JavaScript) | Real-time IoT dashboard |
| Microsoft Excel / CSV | Storage of the sensor dataset for AI training |

## RESULTS AND SYSTEM OUTPUT

Fig. 3 shows the dashboard output of the prototype. The page opens with an overall status banner for the storage condition. Below it, the current temperature, humidity, gas level and moisture readings are shown as separate cards with level bars. A Condition Analysis panel reports the state of the monitored parameters, and a Recommendations panel suggests the action to be taken. The Sensor Details section shows the raw MQ-135 value, the MQ-135 digital state and the raw moisture value, and the System Status section shows the operating mode and the state of the green, yellow and red LEDs and the buzzer.

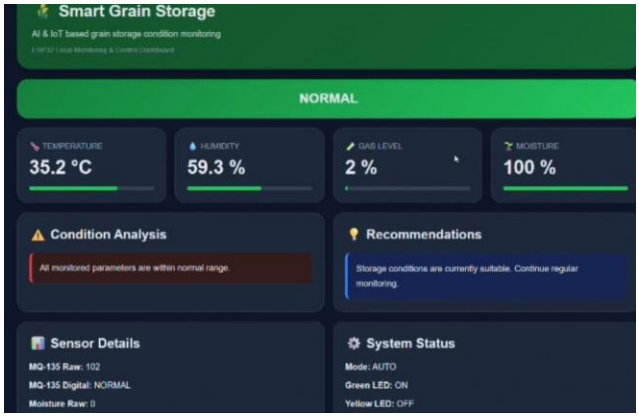



*Fig. 3. Dashboard output of the Smart Grain Storage prototype showing the sensor readings, condition analysis and recommendation.*

In the captured output the status is Normal, the analysis states that all monitored parameters are within the normal range, and the recommendation is to continue regular monitoring. The dashboard also has a System Control section in which the operating mode can be switched between automatic and manual, and the status LEDs can be controlled with on and off buttons. The output demonstrates the complete path from sensing to display: the sensors are read by the ESP32, the readings reach the dashboard, and the system reports the storage condition together with a recommendation. Quantitative performance measures of the AI model, such as classification accuracy on a larger set of storage data, are not reported in this paper.

## FUTURE ENHANCEMENT

The Smart Grain Storage system can be extended with further features to improve monitoring and warehouse management. A mobile application can be developed for real-time monitoring from anywhere, and the sensor data can be stored in cloud storage for long-term analysis and report generation. GSM/SMS and notification alerts can be added for farmers and warehouse managers. A camera-based grain inspection system can be integrated to detect fungal growth and damaged grains. Multiple grain storage units can be monitored from a single dashboard, and data history with graphical reports can be added to analyse storage conditions over time.

## CONCLUSION

This paper presented a Smart Grain Storage Condition Monitoring and Spoilage Risk Prediction System for PDS warehouses using AI and IoT. An ESP32 microcontroller collects temperature, humidity, grain moisture and gas data from low-cost sensors and sends it over Wi-Fi to a cloud server and a dashboard, where an AI model classifies the storage condition into Safe, Moderate Risk and High Risk levels. Alerts through the LCD, LED indicators, buzzer and dashboard allow warehouse staff to act before the grain is spoiled, and the prototype reduces the need for manual inspection.

The literature shows that many existing systems either monitor conditions without predicting risk or have been tested mainly on corn in controlled setups. The present work addresses this gap with a low-cost prototype aimed at PDS warehouses. Validation on larger, multi-grain data from real warehouses, together with the enhancements described above, is the natural next step.

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